I. SUPPLEMENTARY FIGURES

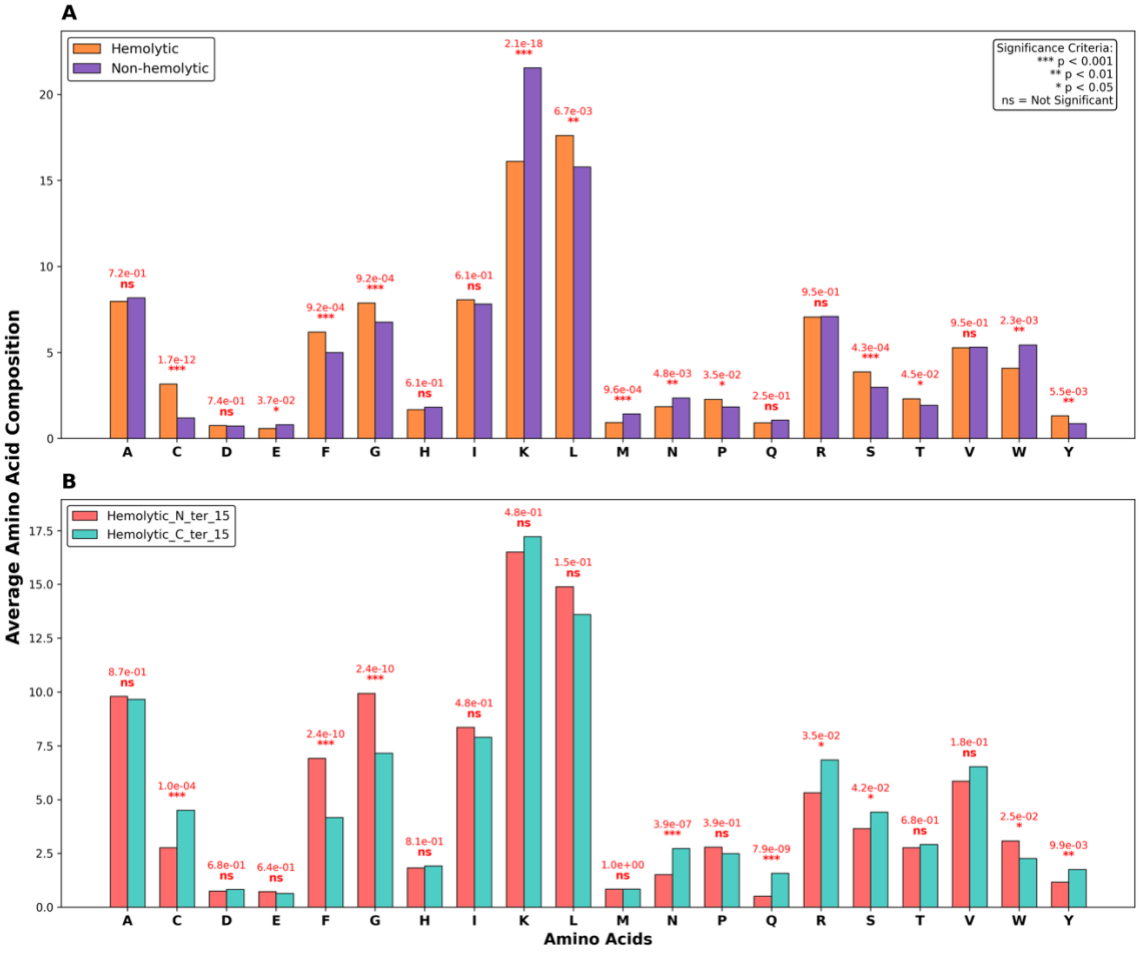


Fig. S1. Amino acid composition analysis. (A) Comparison of amino acid composition between HPs and non-HPs. (B) N-terminal and C-terminal amino acid preferences in HPs. Statistical significance levels: ***p < 0.001, **p < 0.01, *p < 0.05, ns = not significant.

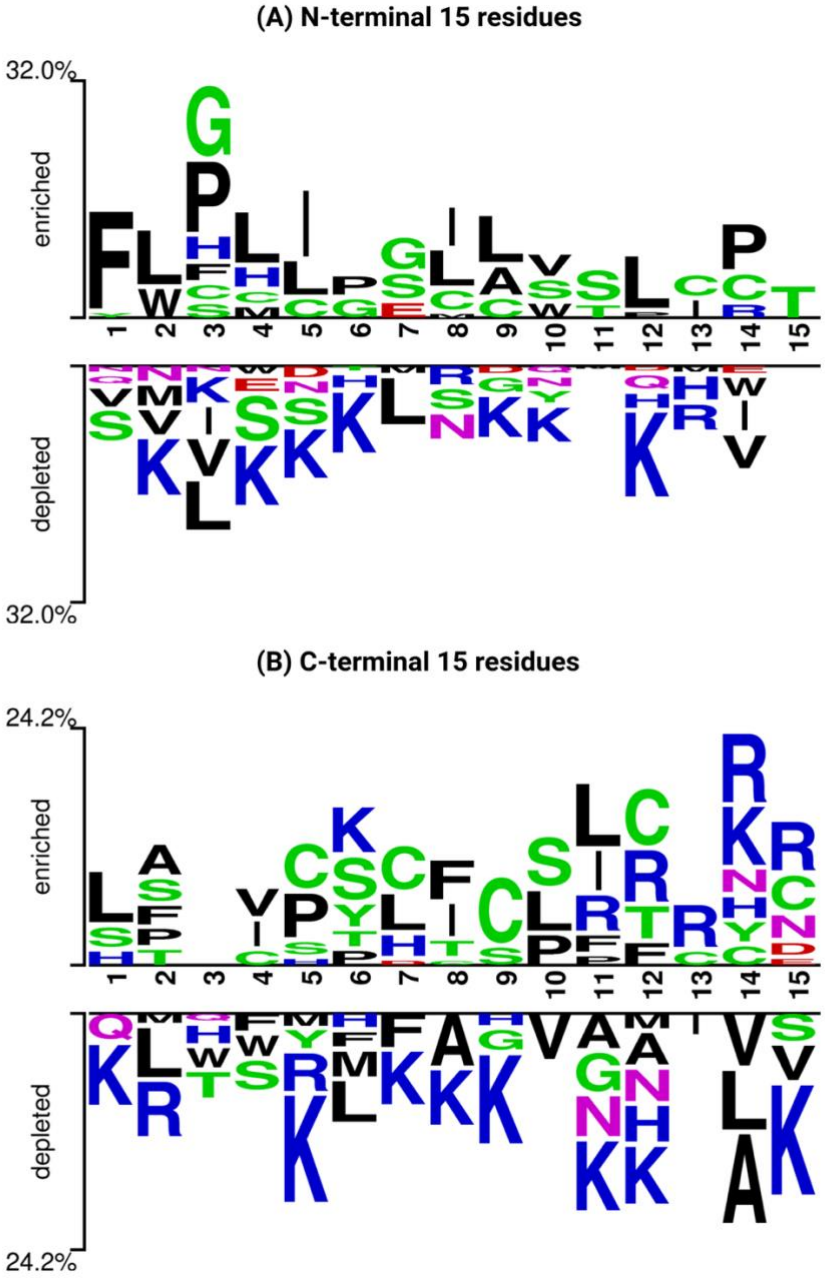


Fig. S2. The Two sample logo (TSL) analysis of positional amino acid preferences in HPs. (A) N-terminal 15 residues and (B) C-terminal 15 residues. Residues above the baseline are enriched in HPs; residues below are depleted.

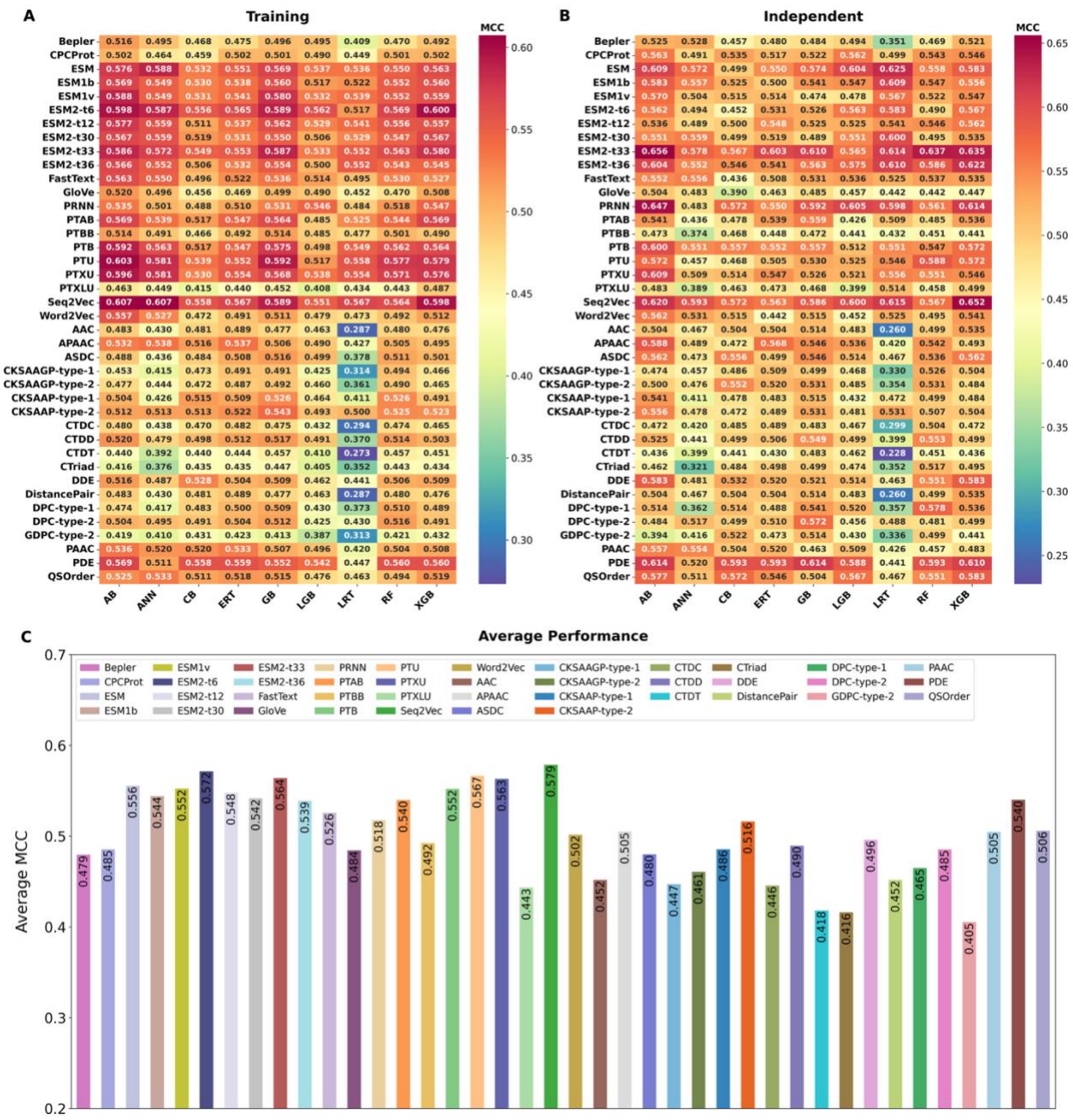


Fig. S3. Classification performance of nine ML classifiers using 21 PLM/NLP-based embeddings and 19 conventional descriptors. Heatmaps showing MCC on the training (A) and independent (B) datasets, and average MCC across the nine classifiers for each descriptor (C).

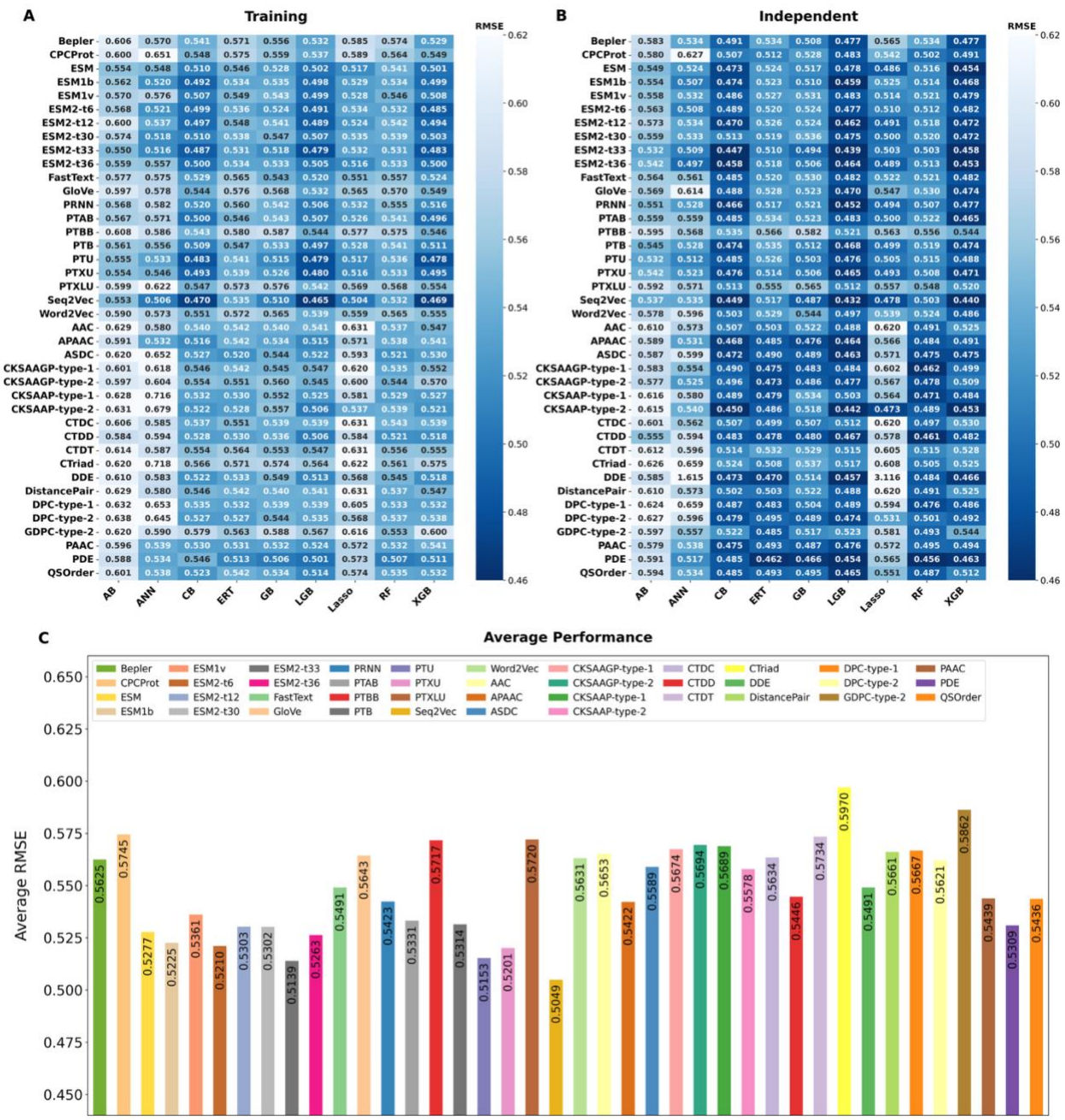


Fig. S4. Performance comparison in terms of root mean square error (RMSE) for nine ML-based regressors using 21 PLM/NLP-based embeddings and 19 conventional descriptors on the training (A) and independent datasets (B). Average RMSE for each feature type (C).

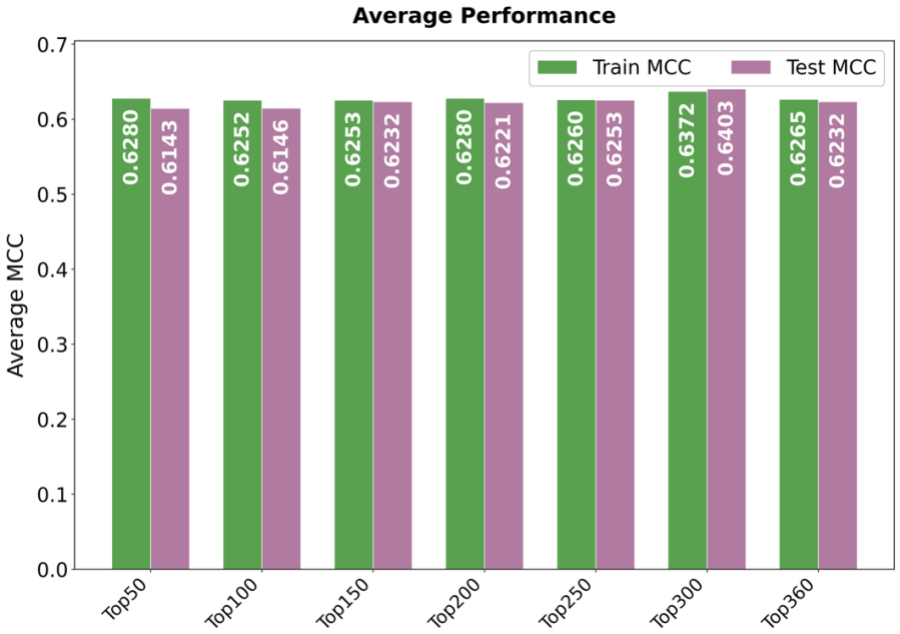


Fig. S5. Performance of classification meta-models across different meta-feature configurations. Bar plots showing the average MCC of classification meta-models developed using Top50, Top100, Top150, Top200, Top250, Top300, and Top360 meta-feature sets on the training and independent datasets.

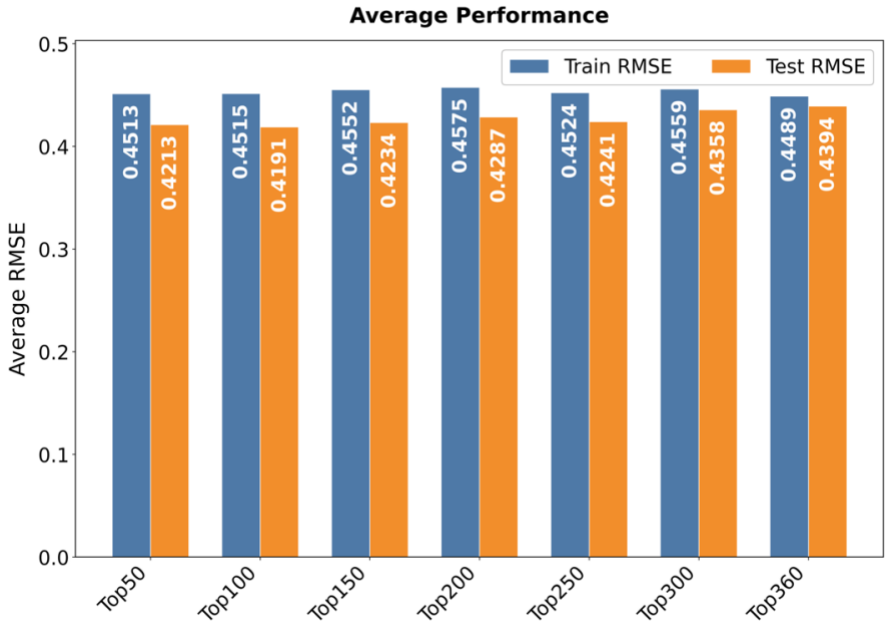


Fig. S6. Performance evaluation of regression meta-models using different meta-feature subsets. The average RMSE values of meta-models generated from different top-ranked baseline model outputs, ranging from Top50 to Top360, are shown for the training and independent datasets.

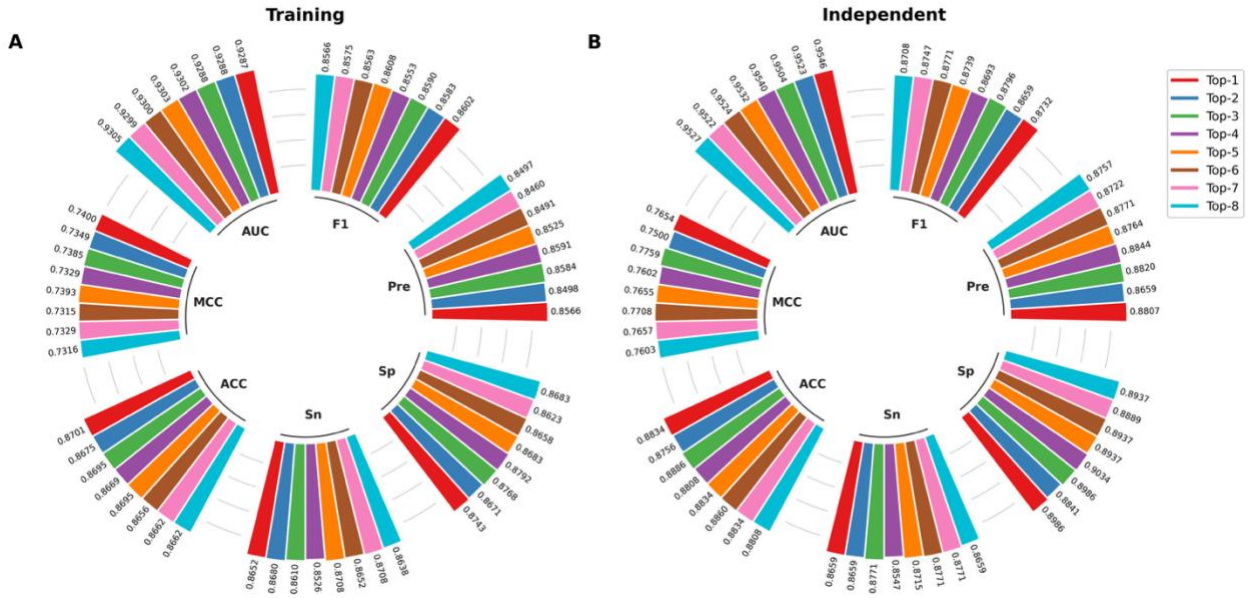


Fig. S7. Classification performance of Top-1 to Top-8 PLM/NLP-based embeddings combined with Top300 Meta-Fe and MERCI-Fe. Circular bar plots showing seven classification metrics on the training (A) and independent (B) datasets.

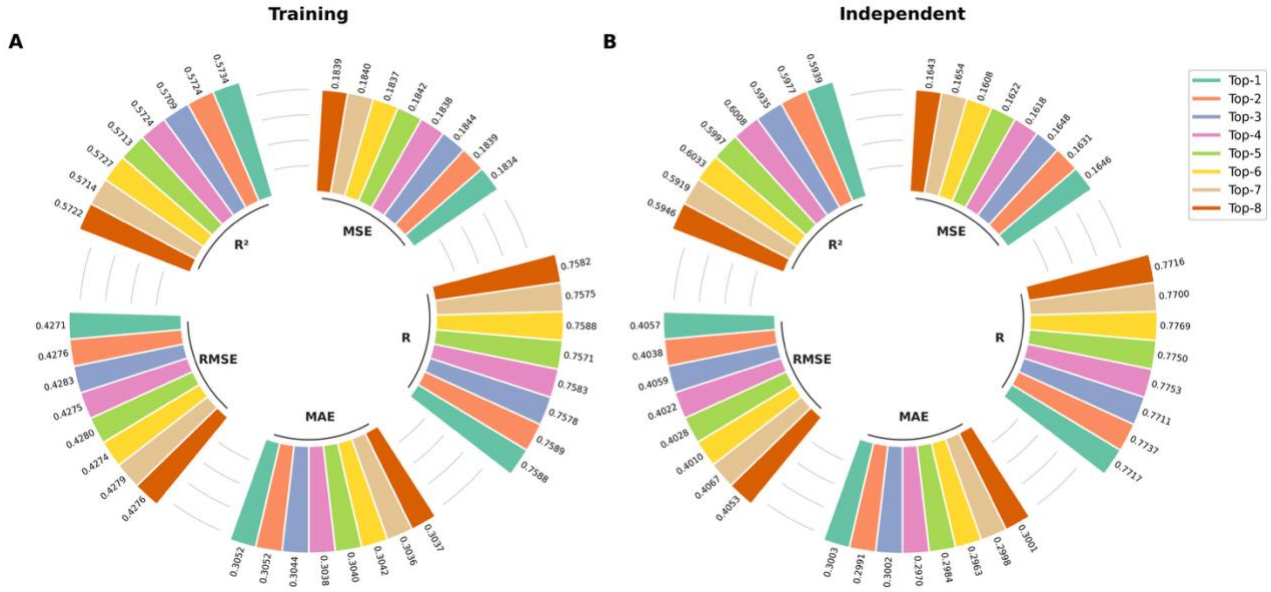


Fig. S8. Regression performance of Top-1 to Top-8 PLM/NLP-based embeddings combined with Top360 Meta-Fe. Circular bar plots showing five regression metrics on the training (A) and independent (B) datasets.

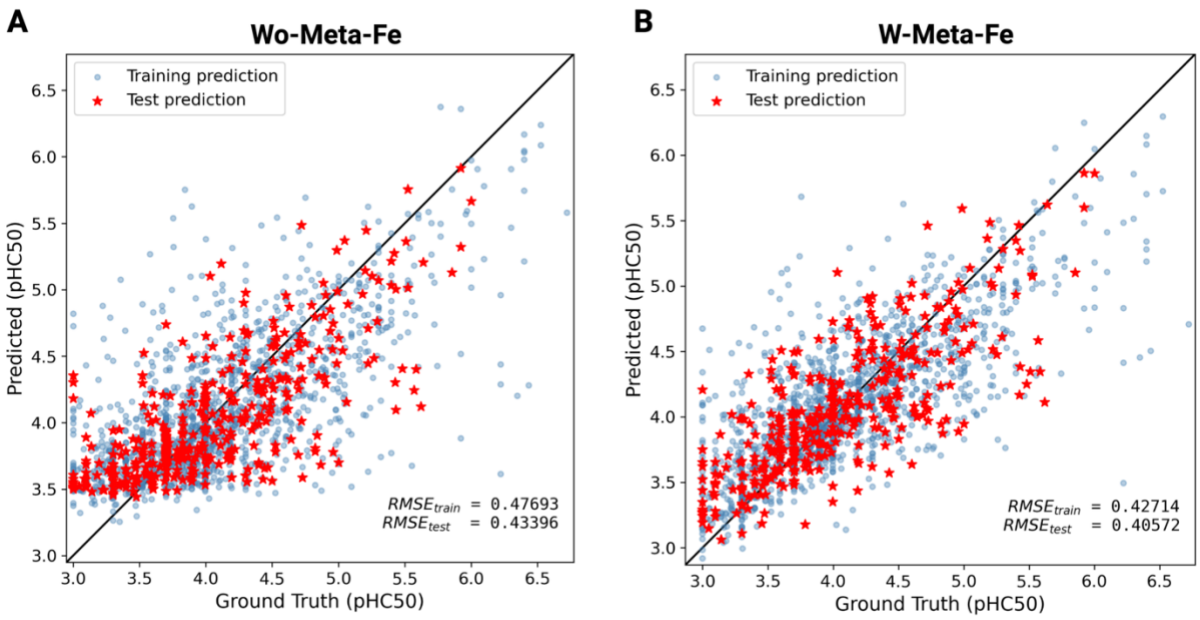


Fig. S9. Regression performance of HLPpred 2.0 with and without Meta-Fe. Scatter plots showing ground-truth versus predicted pHC50 values of models without Meta-Fe (A) and with Meta-Fe (B) on the training and independent datasets. RMSE values are reported in each panel.

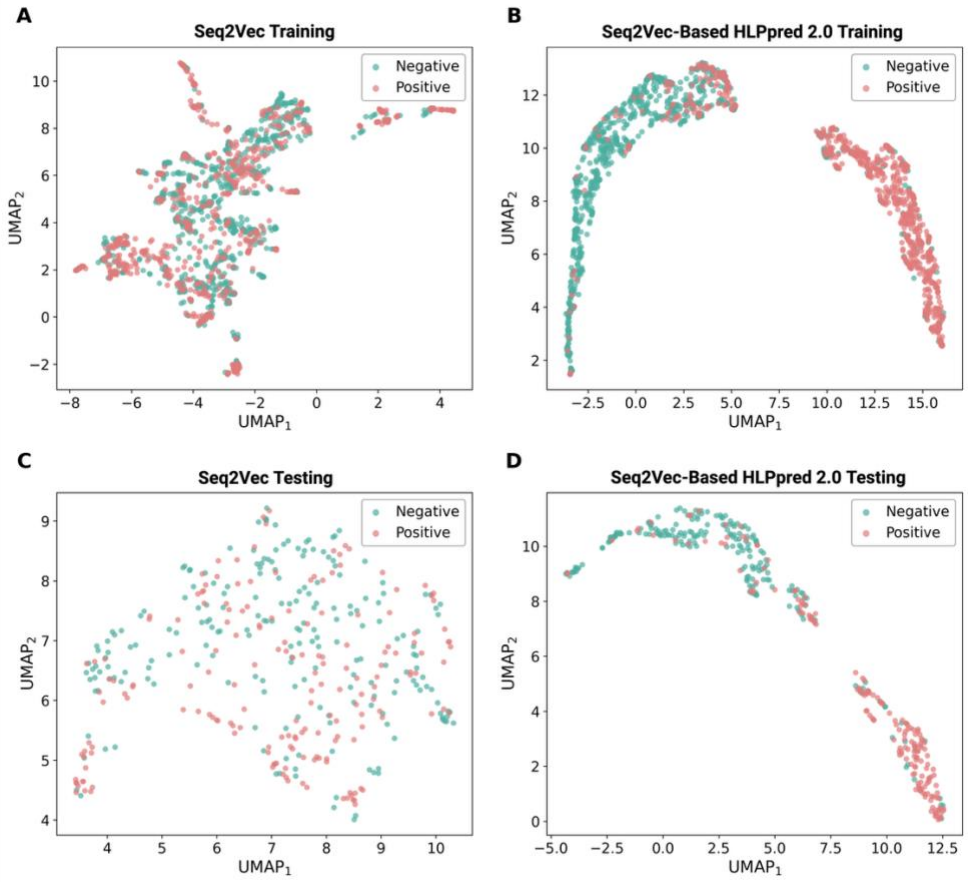


Fig. S10. UMAP visualization of raw mean-pooled Seq2Vec embeddings (A, C) and representations learned by the Seq2Vec-only model (B, D) for the training (A, B) and independent test (C, D) datasets. Salmon and teal indicate HPs and non-HPs, respectively.

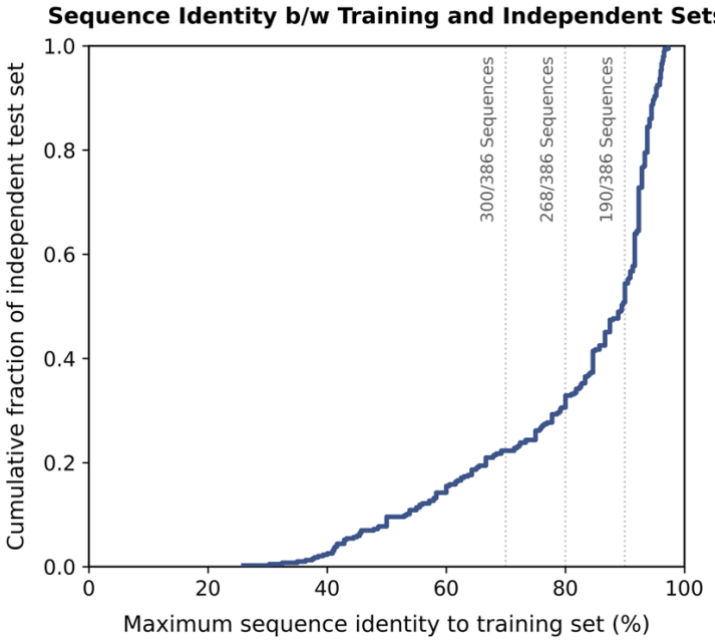


Fig. S11. Sequence identity distribution between the training and independent datasets. Cumulative distribution of the maximum sequence identity of each independent peptide against the training dataset.

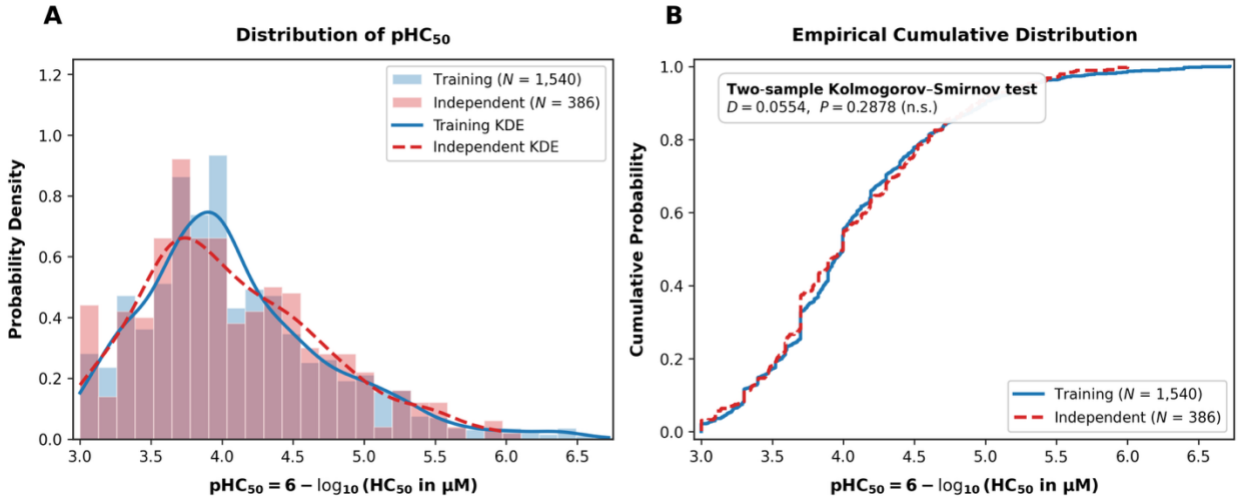


Fig. S12. Comparison of pHC_{50} distributions between the training and independent datasets. (A) pHC_{50} distributions and (B) empirical cumulative distribution functions of the two datasets.

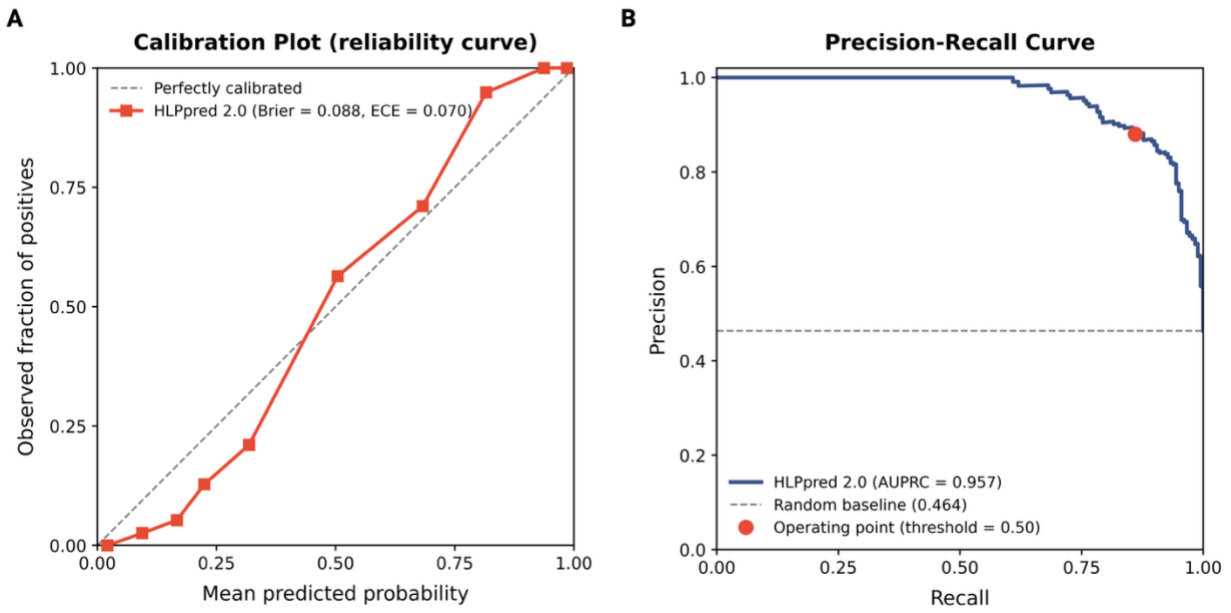


Fig. S13. Probability calibration and precision-recall analysis of HLPpred 2.0. (A) Reliability plot and (B) precision-recall curve on the independent dataset.